GitNotes

Personal Concise Notes

My Personal Concise Notes about how to use Git and GitKraken with GitHub and BitBucket.

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Quick Start

```
                  "Git is a database.
                  Commit is a record."
```

```
git status        # current status of the git directory
git config --list # current config parameters
git log           # show log
        --graph --oneline --all --decorate

git config
        --list
        --global user.name <"full name">
        --global user.email <"email">
        --global core.editor <"command">
                  # eg.: "gedit --new-window --wait"

git init          # create a local repository
git add <file>    # stage <file> for commit and track changes to it
git commit        # commits staged files to local repository
git restore --staged <file> # unstage <file>
git restore --staged .      # unstage all staged files

git rm <list of files>      # delete tracked files
git mv <list of files> <destination>  # move/rename tracked file(s)

git remote add origin <url> # add url of existing remote repository
git push origin master      # push <branch> to <remote>

git branch <branchname>     # (create and) switch to branch
```

Table of Contents

Initializing a Remote Repository on GitHub

We used to be able to create a remote repository from the git command line, but this is no longer allowed. Now, you must either create the repository manually using the web interface, or by using GitHub CLI which accesses GitHub's remote API. This chapter explains how to initialize a new remote Git repository on GitHub from the terminal command line on your development machine.

The GitHub CLI Interface

Token / SSH Keys

You should already be in possession of the token or SSH keys necessary to access either your GitHub account or the repository on which you are working.

Install gh

Install gh.

For example, in Windows you could use PowerShell or download the MSI, and in Linux you can use apt/dpkg, yum/rpm, etc.

Authorize

This is how you authorize gh to access GitHub from the command line using gh:

```
gh auth login
→ github.com
→ https or ssh
    - https
        → paste token
            - token must have minimum scope
                repo
                read:org
                workflow
            - also useful:
                delete_repo
    - ssh
        - you're done
```

Create Remote Repo From Command Line

Basically:

gh repo create <reponame> --public --source=. --remote=origin

If you do not already have your local repository initialized, the full procedure would be:

```
mkdir <reponame>
cd <reponame>
git init
git branch -M master
echo "# <reponame>" >> README.md
git add README.md
git commit -m "Initial commit."
gh auth login
gh repo create <reponame> --public --source=. --remote=origi
# git remote add origin https://github.com/<user>/<reponame>
git push -u origin master
```

Pushing An Existing Local Repo To An Existing Remote Repo

```
git remote add origin https://github.com/<user>/<repo>.git
git branch -M master
git push -u origin master
```

* <user> is your username on GitHub; <repo> is the full reponame on GitHub.

GitHub Markdown Language for .md Files

Some of this is apparently inaccurate. At the moment, GitHub's ***Basic writing and formatting syntax*** page which documents GitHub-Flavoured Markdown (GFM) is located at:

https://docs.github.com/en/get-started/writing-on-github/getting-started-with-writing-and-formatting-on-github/basic-writing-and-formatting-syntax

Comment	<!-- wrap text →
Heading	# h1, ## h2, ### h3, etc.
Issue	#issue-number or #pull-request (to automatically link)
Bold	**bold text**
Italic	*italicised text* or _italicised text_
Bold-Italic	***bold italicised text***
Blockquote	> quoted text
Ordered List	1. first item; 2. second item, etc.
Unordered List	- first item or * first item, etc.
Inline Code	`code`
Code Block	``` fenced code block ```
Horizontal Rule	~~~ or ***
Link	[anchor](https://url.tld)
Image	`![alt text](image.jpg)`
Mention	@username (to automatically link)
Footnotes	Sentence with a footnote.[^1] [^1]: This is the footnote.

Headings can have custom ids:

My Great Heading {#custom-id}

Tables:

```
|Heading1|Heading2|
|---|---|
|Header|Title|
|Paragraph|Text|

minimum width:    ----- (number of dashes)
left justify:     :---
right justify     ---:
fully justify     :---:
```

Definition List

term

: definition

Strikethrough

~~~struckthrough text~~~
~~~

Task List

- [x] completed task

- [ ] incomplete task

Subscript

H~2~O

Superscript

x^2

Callout / Alert

> [!NOTE] / [!TIP] / [!IMPORTANT] / [!WARNING] / [!CAUTION]

> useful information to know

Diagrams and Math

GitHub supports creating diagrams using Mermaid syntax.

Collapsed Section

<details>

<summary>Place summary here.</summary>

text, images, code blocks, headings, etc.

BitBucket

Don't use BitBucket. BitBucket was purchased by Atlassian, so you are really creating an Atlassian account to which you are attaching a free BitBucket subscription. Under this new regime I found repositories and workspaces to be difficult to manage and delete. You cannot delete your Atlassian account while you are subscribed to BitBucket, and you cannot unsubscribe from BitBucket because it is a free subscription.

I also spent way too much time trying to get a push to work from the command line, much less trying to integrate BitBucket with GitKraken. It's an inconvenience up with which one need not put when one has access to GitHub.

Index